RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824–#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

RA8P1 IO allocation

File: mcu_interfaces.kicad_sch

RA8P1 internal core regulator

File: mcu_core_power.kicad_sch

ESP32-C6 radio

RADIO_PWR_EN

File: radio_esp32.kicad_sch

RA8P1 IO and analog supplies

File: mcu_supply_power.kicad_sch

RA8P1 clocks, reset and debug

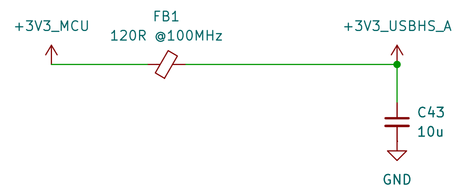
File: mcu_clocks_debug.kicad_sch

Sheet: /
File: ereader_rev1.kicad_sch

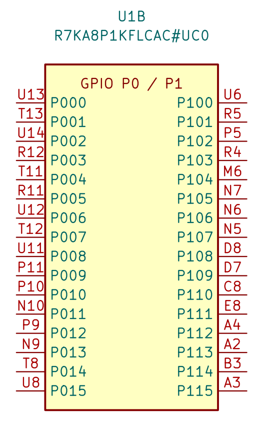
Title: RA8P1 e-reader — schematic index

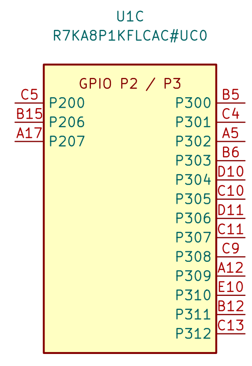
Size: A3	Date: 2026-09-05	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.5		Id: 1/6

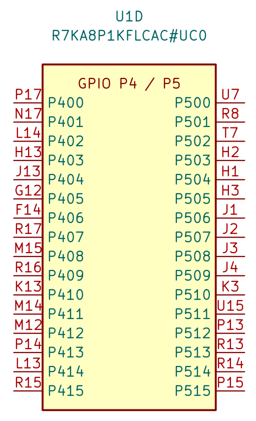
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

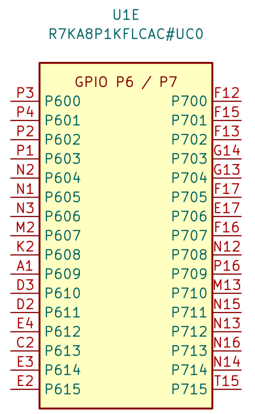


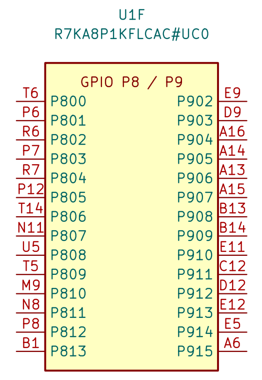
USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = I*R = 2.765 mV; P = I^2*R = 0.153 mW.
After-test DCR 0.10 ohm: 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R; initial 9..11 uF.
TDK nominal bias curve: 9.644 uF at 3.3 V.
Not transient or all-corners qualification; verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]

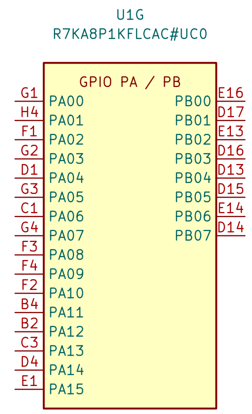


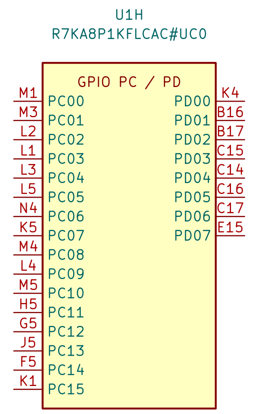


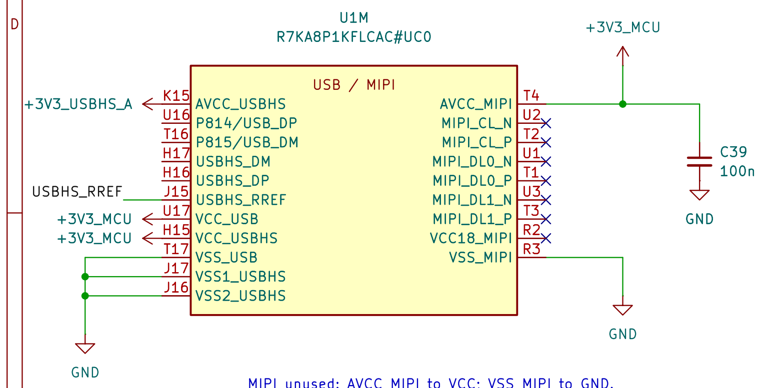


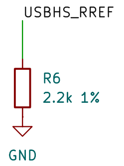


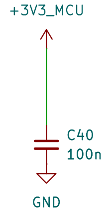


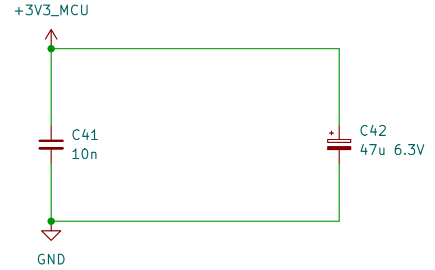












MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.

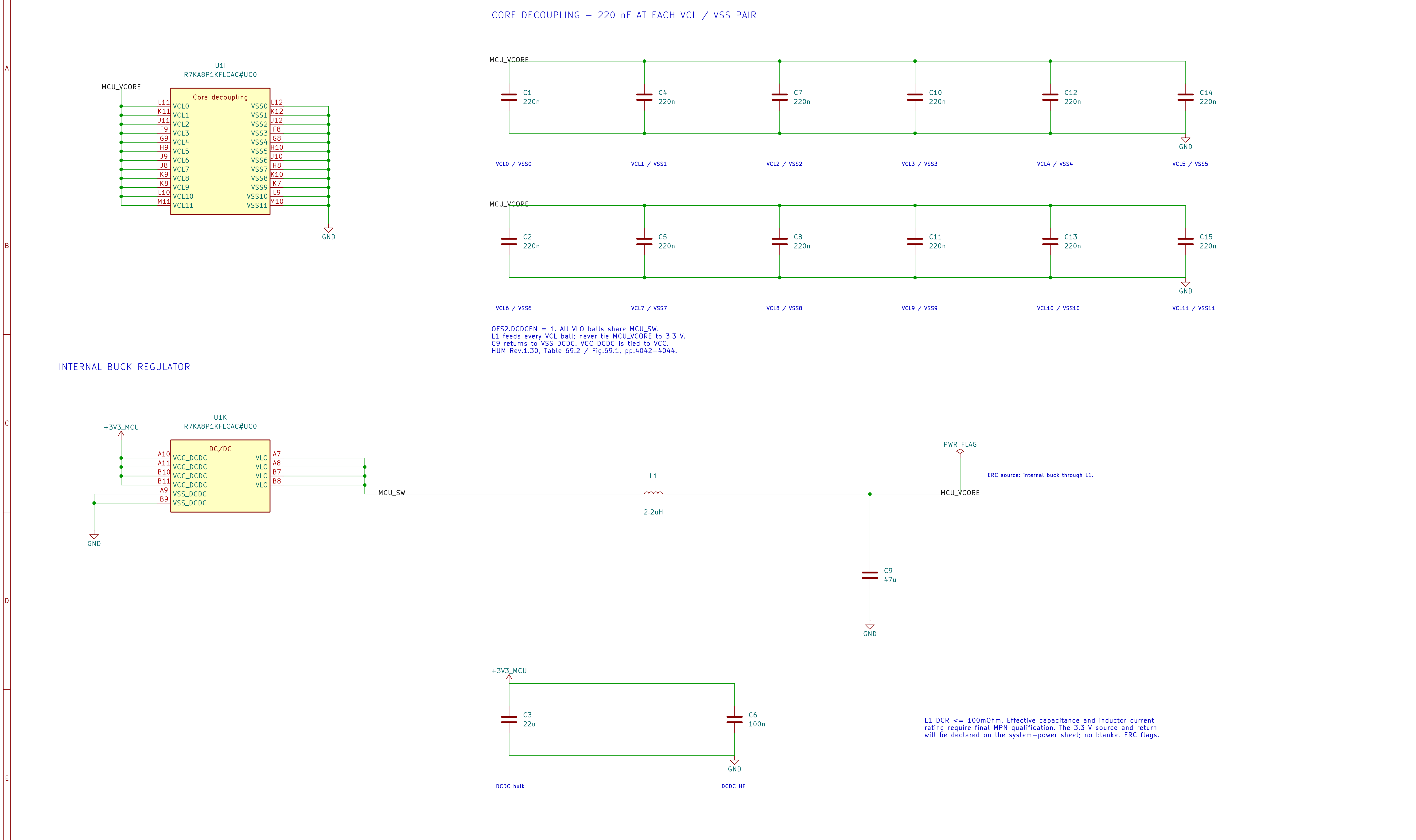
USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1-0.01) = 2178 ohm
Rmax = 2200*(1+0.01) = 2222 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md

USB-002: USB full-speed supply / U1.U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90..110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md

USB-003: USB-HS supply / U1.H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1+/-0.10) = 9..11 nF initial; 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1+/-0.20) = 37.6..56.4 uF initial; 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14% rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

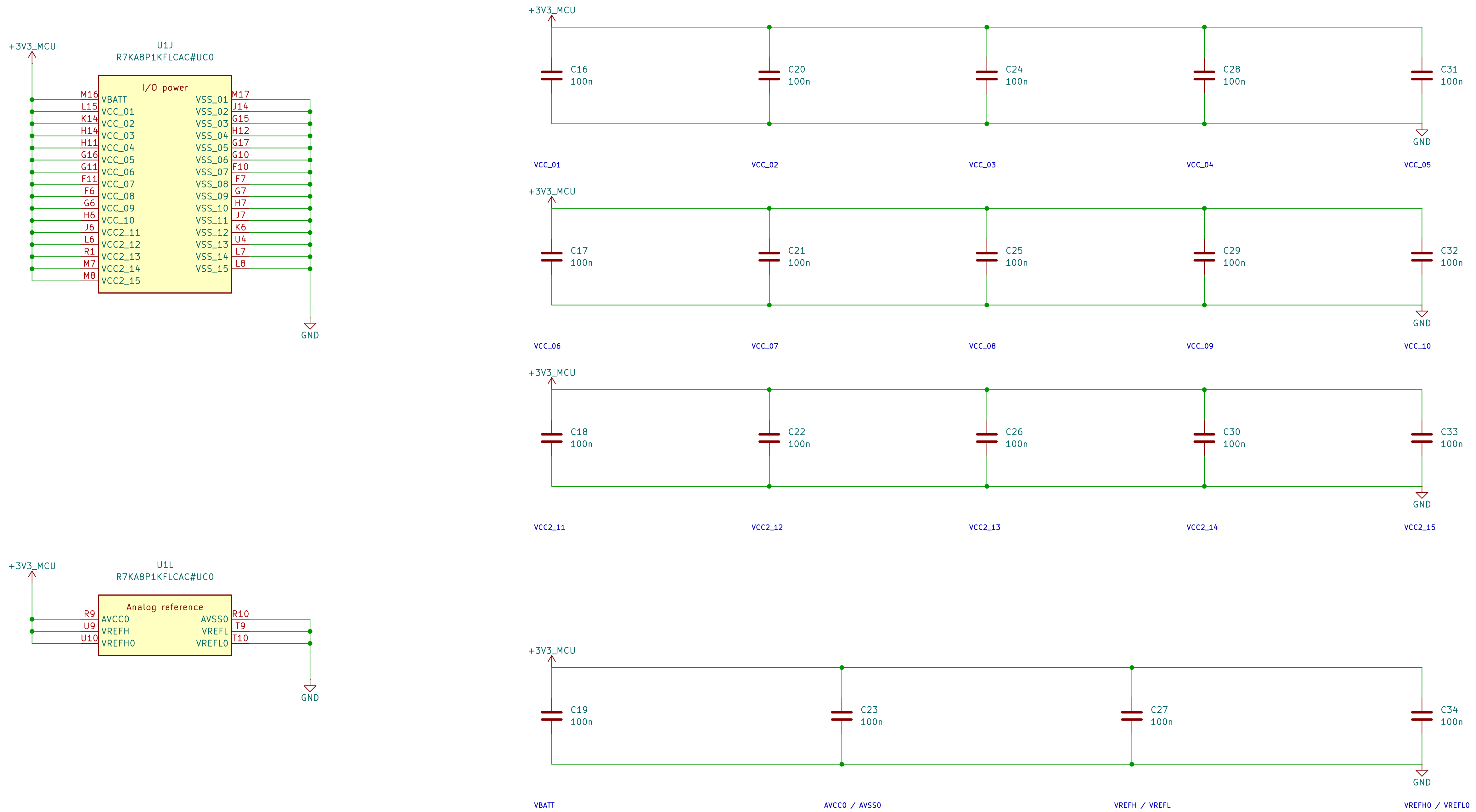
PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90..110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

RA8P1 e-reader	
Sheet: /RA8P1 IO allocation/	
File: mcu_interfaces.kicad_sch	
Title: RA8P1 IO allocation	
Size: A3 Date: 2026-09-05	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.5	Id: 2/6



RA8P1 e-reader
Sheet: /RA8P1 internal core regulator/
File: mcu_core_power.kicad_sch
Title: RA8P1 internal core regulator
Size: A3 Date: 2026-09-05 **Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.5 Id: 3/6

I/O DECOUPLING — 100 nF AT EACH NUMBERED VCC / VSS PAIR

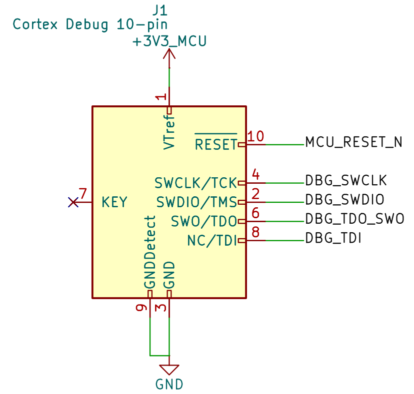


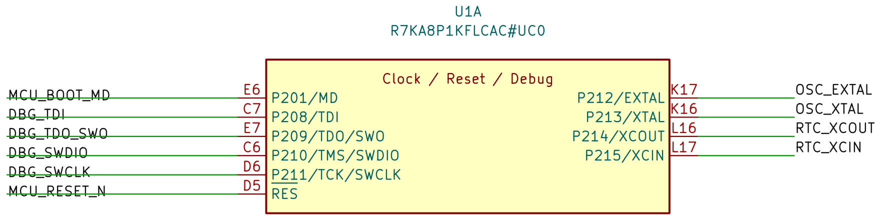
VCC and VCC2 use 3.3 V. All returns use one board ground plane.
VBATT follows the main rail; no separate backup-cell supply.
Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs.
USB/MIPI supplies remain on the interface sheet and are not yet wired.
HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861–862.
This is an electrical draft; capacitor MPN qualification remains open.

RA8P1 e-reader
Sheet: /RA8P1 IO and analog supplies/
File: mcu_supply_power.kicad_sch

Title: RA8P1 I/O and analog supplies

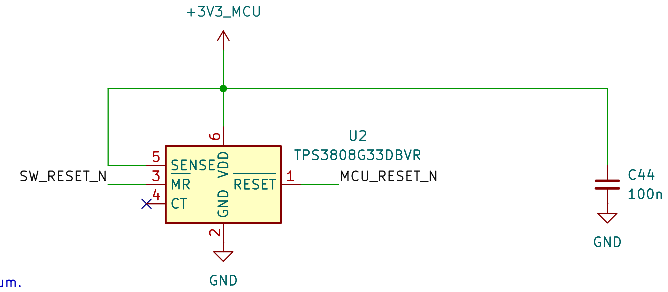
Size: A3 | Date: 2026-09-05 | **Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.5 | Id: 4/6





CLK-001: Y1 24 MHz / Murata XRCGB24M000F3M19R0
Crystal CL = 6 pF; effective Cstray = 2 pF ASSUMED.
CL = C35*C36/(C35+C36) + Cstray
C35 = C36 = 2*(6-2) = 8 pF (NP0, +/-0.5 pF).
Check: 8*8/(8+8) + 2 = 6 pF. At fixed stray: 5.75..6.25 pF.
Different crystal or changed parasitics: recalculate and rematch.
Initial matching only: verify frequency, startup and drive <= 300 uW.
Full derivation + Python check: ../design/clock_calculations.md

CLK-002: Y2 ABS07-LR-32.768KHZ-6-1-T
Crystal CL=6 pF; Cstray=4 pF ASSUMED.
CL=C37*C38/(C37+C38)+Cstray
C37=C38=2*(6-4)=4 pF, +/-0.25 pF NP0.
Check: 4*4/(4+4)+4=6 pF.
Fixed-stray tolerance: 5.875..6.125 pF.
Initial mode: SODRV=10b; SOSEL=0.
Qualify startup, frequency, drive <=0.5 uW.
Full math/Python: ../design/clock_calculations.md



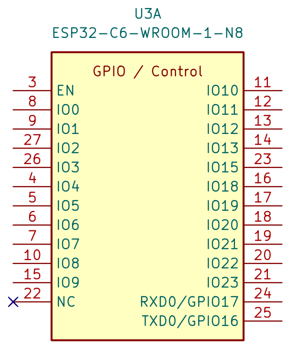
RST-001: U2 supervises +3V3_MCU; SW1 asserts MR.
CT open: release delay 12..28 ms; 12/2.4 = 5x MCU minimum.
Falling threshold = 3.07*(1 +/- 0.015) = 3.02395..3.11605 V.
Conservative release screen = 3.11605*1.025 = 3.19395125 V.
ASSUMED rail +/-2%: 3.3*0.98 - 3.19395125 = 40.05 mV margin.
R1 worst pull-up current = 3.6/9900 = 0.364 mA < 1 mA.
RES low margin = 0.2*3.02395 - 0.4 = 0.20479 V.
No RES capacitor; open-drain reset drivers only.
External RES startup path; do not assume internal POR flag.
Full math, limits and Python: design/boot_and_debug.md RST-001.

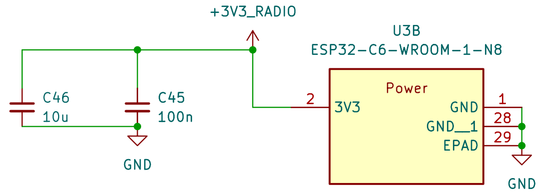
References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3-4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.

Sheet: /RA8P1 clocks, reset and debug/
File: mcu_clocks_debug.kicad_sch

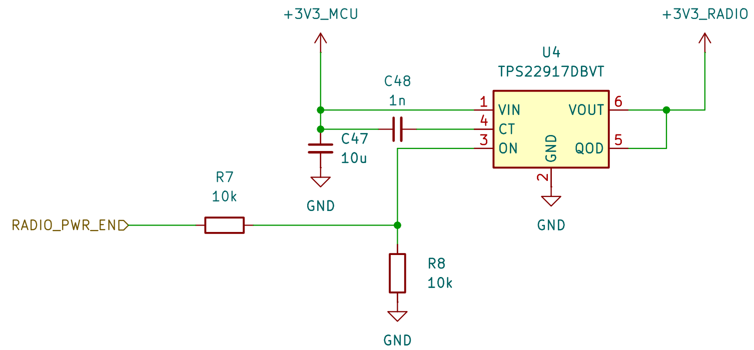
Title: RA8P1 clocks, reset and debug

Size: A3	Date: 2026-09-05	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.5		Id: 5/6





RADIO-003: C46 bulk bypass / TDK C3216X7R1V106K160AC
10 uF nominal, +/-10%, X7R; DC bias reduces capacitance.
C(3.3 V) = 9.69875 + (3.3-3.15)/0.85*(9.38950-9.69875)
= 9.64418 uF; C(3.6 V) = 9.53503 uF (nominal).
Source curve interpolation, NOT a guaranteed minimum.
Rail: 3.0..3.6 V; source capability >= 0.5 A.
Supply and transmit transient qualification remain required.
Calculations: ../design/radio_interface.md#radio-003-bulk-capacitance



RADIO-004: U4 TPS22917DBVT / radio power gating
QOD tied to VOUT: Rdis = 150 ohm TYPICAL.
C45+C46 = 0.1+10 = 10.1 uF nominal external load.
t90..10 = ln(9)*150*10.1u = 3.329 ms (ideal RC).
Not a guaranteed off-time: module C/load and VIN decay matter.
C48 connects CT to VIN, not GND; 50 V rating >=7 V required.
Host control, reset/isolation and upstream regulator remain incomplete.
Math + limits: ../design/radio_interface.md#radio-004-radio-load-switch

RADIO-005: C47 / U4 VIN local bypass
10 uF nominal, +/-10%, 35 V X7R; same exact TDK part as C46.
C(3.3 V) = 9.69875 + (3.3-3.15)/0.85*(9.38950-9.69875)
= 9.64418 uF nominal; NOT a guaranteed minimum.
Droop = integral(Iload-Isource)dt / Ceff + ESR/ESL effects.
TI section 11: 1 uF usually sufficient; slow sources may need more.
C47 is upstream: exclude it from C45+C46 output-discharge math.
Regulator/transient qualification required; no source flag added.
Calculations: ../design/radio_interface.md#radio-005-input-bypass

RADIO-006: R7/R8 / active-high enable, default OFF
R7 series = R8 pulldown = 10k 1%; +/-2% screening envelope.
TPS22917 ON: low <=0.35 V; high >=1.0 V.
Assume node leakage <=10 uA and host VOL<=0.5 V, VOH>=2.5 V.
Rparallel <=5.1k; divider fraction = 0.49..0.51.
Vlow <=0.5*0.51+10u*5.1k = 0.306 V <0.35 V.
Vhigh >=2.5*0.49-10u*5.1k = 1.174 V >1.0 V.
Hi-Z screen: 10u*10.2k = 0.102 V; disable host pull-up.
Host pin, all power states, leakage and timing need qualification.
Full math: ../design/radio_interface.md#radio-006-enable-divider

RADIO-007: C48 / CT slew-rate control, TDK C1608NP01H102J080AA
1nF +/-5%, 50 V NP0; CT-to-VIN, NOT CT-to-GND.
TI 3.6 V / 25 C TYPICAL table point, not guaranteed at 3.3 V:
Slew = 1900/1000 = 1.9 mV/us; tON = 3.8*1000 = 3.8 ms.
tR(10..90%) = 1.6*1000 = 1.6 ms (TI test load).
C45+C46 = 10.1 uF nominal: Icap = C*dV/dt = 19.19 mA.
Open-CT comparison: 10.1u*44mV/us = 444.4 mA (typical).
Total inrush also includes module C/load and future rail parts.
Not a current limiter or reset delay; startup qualification required.
Full math: ../design/radio_interface.md#radio-007-slew-rate-capacitor

Incomplete electrical design - not for manufacture

Sheet: /ESP32-C6 radio/
File: radio_esp32.kicad_sch

Title: ESP32-C6 radio

Size: A3	Date: 2026-09-05	Rev: DRAFT
KiCad E.D.A. 10.0.5		Id: 6/6